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UTHM INDUSTRIAL VISIT

NEWSLETTER

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SUMMARY

On 3 December 2025, Universiti Teknologi Malaysia (UTM) conducted an industrial visit to Universiti Tun Hussein Onn Malaysia (UTHM) to observe their newly constructed Digital Gateway hub. The goal was to demonstrate how fundamental system design concepts are applied to meet the demands of a large-scale institution. During the visit, students examined the command operations centre and utility rooms, receiving technical briefings on the facility's cooling systems, containment technologies, and security protocols. The session concluded with a Q&A session with the IT management team, which provided critical insights into balancing legacy hardware maintenance with modern data security requirements.

The Technologies and Issues

During the visit, UTHM introduced several technologies used to support their ICT operations. Their data centre relies entirely on on-premise infrastructure, with cold and hot aisle containment to maintain efficient cooling and stable server performance. All student information is stored in a single master database, and data is transferred to other systems through self-developed APIs, Single Sign-On (SSO), and occasional manual updates. The university receives few major complaints, with the most common issues involving forgotten passwords, SSO login errors, and campus WiFi connectivity. To manage peak system load, especially during course registration, UTHM handles around 1,000 concurrent users by separating registration dates by student year. They are also migrating from an older client-server system to a fully web-based platform, a process that is now 70–80% complete. In terms of cybersecurity, UTHM protects its systems with two firewalls from different regions, a Web Application Firewall (WAF), and a dedicated security committee that responds quickly to potential threats.



LAU YEE WEN A23CS0099

The industrial visit to UTHM helped me understand how a real data centre operates and how Enterprise Architecture (EA) guides strategic IT planning. Seeing the hot/cold aisle layout showed me how the university manages cooling, cost, and reliability effectively. I also learned how the master database acts as the single source of truth and the challenges of integrating systems like the LMS. Overall, the visit showed me that digital transformation requires not just new technology but careful, phased planning to maintain a smooth user experience.

CHERYL CHEONG KAH VOON A23CS0060

From this visit, I learned how UTHM manages its ICT infrastructure, especially through a single master database and system integration using APIs and SSO. The explanation on system migration helped me understand the challenges of upgrading large-scale platforms. Seeing the data centre's cooling setup and security measures also gave me a clearer picture of how real organisations protect and maintain their systems. Overall, the visit strengthened my understanding of enterprise architecture in practice.